
Lab 1

Lab 2: Workflow with PostgreSQL

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

Academic Year: 2024–2025

Objective

Guide students to set up PostgreSQL, create a user and database, load initial lab tables, perform CRUD operations, and save all successfully run commands.

Lab Workflow

1. Open PostgreSQL Environment

- Terminal: Open a terminal and enter PostgreSQL interactive shell:
`psql`
- Or, pgAdmin: Open pgAdmin and connect to the PostgreSQL server.

2. Drop Existing Database and User (if they exist)

- Terminal:
 - (a) Open terminal and enter `psql`.
 - (b) Run the following commands to safely drop database and user:

```
DO $$
BEGIN
    IF EXISTS (SELECT FROM pg_database WHERE datname = 'cse4308_12345') THEN
        EXECUTE 'DROP DATABASE cse4308_12345';
    END IF;
END$$;

DO $$
BEGIN
    IF EXISTS (SELECT FROM pg_roles WHERE rolname = 'alice') THEN
        EXECUTE 'DROP ROLE alice';
    END IF;
END$$;
```
 - (c) Here, use your name in small letters for username instead of "alice" and your full_id(230041101) instead of "12345" for database name
- Or, pgAdmin Alternative:
 - Right-click database `cse4308_12345` → Drop
 - Navigate to Login Roles → Right-click user `alice` → Drop

3. Create User and Database

- Terminal:
 - (a) Enter `psql`.
 - (b) Run:

```
CREATE ROLE alice LOGIN;  
CREATE DATABASE cse4308_12345 OWNER alice;  
GRANT ALL PRIVILEGES ON DATABASE cse4308_12345 TO alice;
```

- pgAdmin Alternative:
 - Login Roles → Create → Name: `alice`, Enable Login
 - Databases → Create → Name: `cse4308_12345`, Owner: `alice`
 - Privileges are automatically granted to owner

4. Connect to the Database as the User

- Terminal:

```
psql -d cse4308_12345 -U alice
```
- pgAdmin:
 - Connect using login `alice` and select database `cse4308_12345`

5. Load Lab 1 SQL File

- Terminal:

```
psql -d cse4308_12345 -U alice -f /path/to/lab1.sql
```
- pgAdmin:
 - Open Query Tool → Open File → Select `lab1.sql` → Execute (F5)

6. Verify Tables and Data

- Terminal:

```
\dt  
SELECT * FROM students;  
SELECT * FROM courses;
```
- pgAdmin: Expand Tables → View data

7. Perform CRUD Operations on Lab 1 Data

- CREATE:

```
INSERT INTO students (student_id, name, major, gpa)  
VALUES (104, 'David', 'ME', 3.60);  
  
INSERT INTO courses (course_id, course_name, credits)  
VALUES (4, 'Machine Learning', 3);
```
- READ:

```
SELECT * FROM students WHERE major='CSE';
SELECT * FROM courses WHERE credits=3;
SELECT * FROM students ORDER BY gpa DESC;
```

- UPDATE:

```
UPDATE students SET gpa=3.55 WHERE name='Bob';
UPDATE courses SET credits=4 WHERE course_name='Operating Systems';
```

- DELETE:

```
DELETE FROM students WHERE name='Charlie';
DELETE FROM courses WHERE course_name='Digital Electronics';
```

- pgAdmin Alternative: Run the above queries in Query Tool.

8. Save Successfully Run Commands

- File naming convention: fullid_CSE4308_L1_T1.sql
- Example: 12345_CSE4308_L1_T1.sql
- Save all successfully executed commands including table creation, inserts, and CRUD operations

9. Organize Lab Directory for Submission

```
/CSE4308_Labs/
  /Lab1/
    lab1.sql
    12345_CSE4308_L1_T1.sql
  /Lab2/
    ...
```

Notes

- All terminal commands are run in `psql`.
- Steps in pgAdmin are alternatives to doing the same actions in terminal.
- CRUD operations must be performed **as the student user** to practice permissions.
- Only successfully executed commands should be saved in `.sql` files.